

SAINT-LOUIS, Senegal

WMO: 616000

Lat: 16.051N

Lon: 16.463W

Elev: 9

StdP: 14.69

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.0	60.7	29.8	23.9	80.5	33.5	28.1	77.7	18.7	76.8	16.5	77.8	5.4	40	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	11.9	101.9	70.8	98.4	70.1	95.2	70.4	83.4	89.4	82.5	88.2	81.8	87.2	8.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.0	166.5	86.8	80.9	160.7	85.7	80.5	158.3	85.3	47.3	89.8	46.4	88.5	45.5	88.1	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	15.3	13.8	DB	53.8	109.1	2.6	1.9	51.9	110.5	50.3	111.6	48.9	112.7	47.0	114.1	(4)
(5)			WB	46.4	85.8	1.9	2.2	45.0	87.5	43.9	88.8	42.8	90.0	41.5	91.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.6	75.0	76.0	77.0	76.3	76.7	79.9	82.5	83.6	84.4	85.1	81.8	77.0
	DBStd	5.21	4.11	4.60	5.61	4.99	4.28	3.11	2.09	1.90	2.31	3.38	3.49	3.92
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10809	773	728	836	788	827	896	1007	1043	1033	1087	955	836
	CDD65	5335	309	308	371	338	362	446	542	578	583	622	505	371
	CDH74	51024	2541	2600	3022	2323	2385	3749	5789	6840	7055	7113	4597	3010
	CDH80	21330	1154	1256	1532	993	801	1099	1871	2640	2997	3257	2329	1401
Wind	WSAvg	7.8	7.4	8.4	9.2	10.0	9.4	8.3	7.9	7.0	6.1	6.7	6.7	6.9
Precipitation	PrecAvg	9.9	0.0	0.0	0.0	0.0	0.0	0.3	1.8	3.7	4.1	0.7	0.0	0.0
	PrecMax	20.9	0.8	0.6	0.2	0.1	0.1	2.8	8.1	9.5	15.0	7.8	0.6	1.3
	PrecMin	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.2	0.0	0.0	0.0
	PrecStd	3.9	0.2	0.1	0.0	0.0	0.0	0.5	1.7	2.0	2.9	1.4	0.1	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.0	100.3	103.8	106.3	102.1	95.9	92.9	93.9	96.4	105.9	103.6	98.7
		MCWB	67.3	67.0	68.4	71.4	70.3	76.2	79.7	80.8	81.1	74.4	70.9	67.8
	2%	DB	93.2	95.9	98.9	95.2	91.9	90.2	89.9	90.9	92.8	99.8	99.2	94.9
		MCWB	65.3	65.9	68.2	69.9	71.9	77.0	79.7	81.2	81.7	73.3	69.9	67.4
	5%	DB	90.2	92.3	94.2	89.4	87.4	87.5	88.2	89.5	91.2	94.7	95.7	91.5
		MCWB	64.4	64.6	66.5	69.8	72.2	77.1	79.3	80.6	81.7	77.2	69.5	66.5
	10%	DB	87.1	88.2	88.6	84.9	83.9	85.6	87.2	88.1	89.7	91.5	92.2	88.0
		MCWB	63.7	63.9	66.3	69.6	72.1	76.7	78.9	80.2	81.2	79.4	70.3	65.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.9	72.6	75.1	75.6	77.6	80.3	82.6	83.5	84.6	84.6	80.4	76.4
		MCDB	88.0	86.8	91.3	93.7	86.3	87.3	88.5	89.8	90.7	91.1	87.9	87.8
	2%	WB	69.9	70.4	72.4	73.1	75.8	79.2	81.0	82.4	83.5	83.2	78.4	73.8
		MCDB	84.2	85.0	88.5	86.7	84.0	85.6	86.9	88.0	89.5	89.3	85.8	83.3
	5%	WB	68.1	68.8	70.5	71.7	74.6	78.3	80.2	81.6	82.7	82.1	76.4	72.0
		MCDB	82.1	83.2	84.5	84.2	82.2	84.4	85.9	87.1	88.2	88.0	84.8	81.6
	10%	WB	66.5	67.3	68.9	70.3	73.2	77.5	79.6	80.9	82.0	81.1	74.8	70.0
		MCDB	79.2	80.5	81.5	81.5	80.7	83.4	85.1	86.2	87.2	86.6	83.1	80.2
Mean Daily Temperature Range	5% DB	MDBR	24.9	24.6	23.2	20.0	16.0	12.0	10.1	10.6	11.9	16.2	23.0	24.4
		MCDBR	29.3	30.4	32.1	27.4	22.8	15.6	11.8	12.2	13.9	22.5	28.4	28.9
		MCWBR	13.1	11.9	10.7	8.9	8.2	5.6	4.8	5.2	6.0	7.8	10.7	12.2
	5% WB	MCDBR	25.4	24.7	24.7	22.2	16.8	12.4	11.0	11.6	12.3	15.1	22.1	24.1
		MCWBR	12.4	10.9	10.1	8.9	7.3	5.2	5.1	5.7	6.2	6.9	9.8	11.8
Clear-Sky Solar Irradiance	taub	0.541	0.556	0.612	0.636	0.703	0.768	0.746	0.675	0.640	0.614	0.518	0.519	
	taud	1.808	1.786	1.686	1.693	1.570	1.530	1.594	1.742	1.801	1.814	1.964	1.913	
	Ebn at Noon	236	241	232	227	209	194	199	215	223	224	242	238	
	Edh at Noon	65	70	79	79	88	91	85	74	70	67	56	57	
All-Sky Solar Radiation	RadAvg	1582	1785	2094	2257	2246	2099	1959	1895	1828	1828	1629	1504	
	RadStd	111	102	72	75	86	50	59	60	77	77	88	67	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page